

From Defect Detection to Flight Clearance: a Bayesian SHM Stack for Reusable Launch Vehicles

Stage 0–3 implemented and verified on CFRP thermoelastic fields

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June 2026

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Expendable: “will it survive *one* flight?” — a single pass/fail margin check.

Reusable: “is this vehicle OK to fly *again*, and how many flights remain?” — a recurring, history-dependent decision.

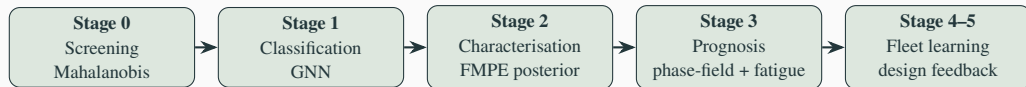
- Same vehicle re-inspected after every flight $\Rightarrow$ *fixed geometry, fixed individual* is the norm, and its own past-flight scans form a growing healthy reference
- SpaceX (empirically): preventive replacement $\rightarrow$ *condition-based monitoring*; staged certification 20 $\rightarrow$ 40 flights; fleet-leader B1067 at 35 flights (06/2026)

- Cost asymmetry: vehicle + payload loss $\gg$

This work

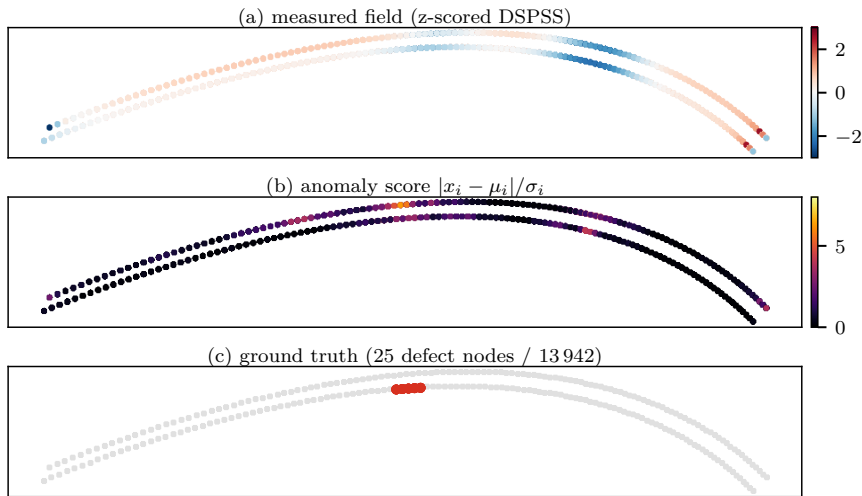
Formalise that practice as a *measurable, Bayesian, physics-based* pipeline — detect $\rightarrow$ characterise (with uncertainty) $\rightarrow$ predict crack growth $\rightarrow$ clear — applicable from the *first* article, not after a decade of flights.

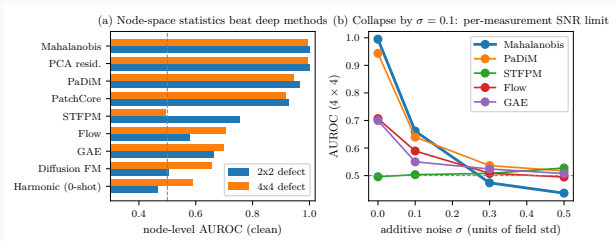
Data: CFRP interstage thermoelastic full-field (DSPSS, 13 942 nodes) + H3 fairing guided waves.



- **Green = implemented and tested** (2026-06; two git repos, all committed; 19/19 tests green)
- Each stage answers one question: **where?** (Stage 0) → **what?** (Stage 1) → **how bad, with what uncertainty?** (Stage 2) → **fly again?** (Stage 3)
- One data contract end-to-end: posterior $\theta = (c_x, c_y, \text{layer}, \log_2 \text{size})$ flows from Stage 2 *directly* into Stage 3's defect seeding
- Stages 0–2 are model-light (statistics / amortised inference); Stage 3 carries the mechanics (phase-field fracture)

What the system sees: detection at a glance





Score $s_i = |x_i - \mu_i|/\sigma_i$, with (μ_i, σ_i) per node from $N=2000$ healthy fields. **Node AUROC (clean):**

- Mahalanobis / PCA: **0.999 / 0.995** (2x2 / 4x4)
- PaDiM 0.96, PatchCore 0.93, STTFPM 0.75, GAE 0.66, diffusion 0.50–0.66

SNR limit (right panel)

All methods collapse by $\sigma=0.1$: defect amplitude $<0.1\sigma$. Lock-in IR thermography averages $\sim 10^3$ cycles ($\sigma \propto 1/\sqrt{K}$) $\Rightarrow$ an SNR *requirement*, not a blocker.

Fixed geometry = template statistics

Re-inspecting the *same* vehicle: healthy manifold is a point cloud around one template.
Per-node (μ_i, σ_i) is the correct inductive bias.

Complementary weaknesses

	Mahalanobis	GNN
unknown defect size	open	collapses
noise (12%)	dies	robust
semantic class	—	only option
new airframe	needs refs	transferable

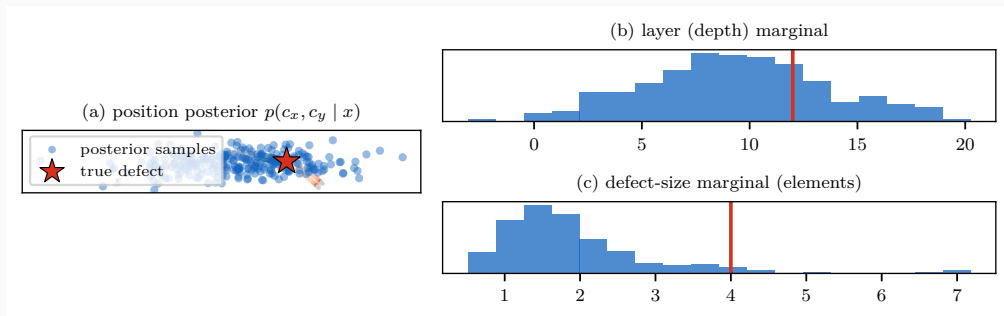
⇒ statistics dominate as a vehicle accumulates flight history; learned models bootstrap *new* airframes and repairs.

Flow-matching posterior estimation, trained on the *existing* FEM set (20k pairs, zero new simulations):

parameter	RMSE	R^2	90%-coverage
position c_x	0.055	0.78	0.944
position c_y	0.021	0.83	0.940
size (3-class)	—	0.73 (acc. 0.82)	0.894
layer (depth)	2.98	0.63	0.870

- Coverage $\approx$ target 0.90 $\Rightarrow$ **calibrated** posteriors, safe to propagate downstream
- Depth weakly identifiable — a physics limit of surface thermoelastic measurement, reported honestly

Stage 2 in pictures: the posterior, not a point estimate



One measured field $\rightarrow$ 256 posterior draws. Position (a) concentrates at the true defect (star); depth (b) is honestly broad — the physics limit of a surface measurement; size (c) shows why we propagate the *whole* posterior into Stage 3 instead of trusting one number.

Energy $\Psi = \int \left[g(d) \psi_e^+ + \frac{G_c}{c_w} \left(\frac{d^2}{\ell} + \ell \nabla d \cdot A \nabla d \right) \right] dV$, with structural tensor $A = I + \beta a \otimes a$ (fiber dir. a).

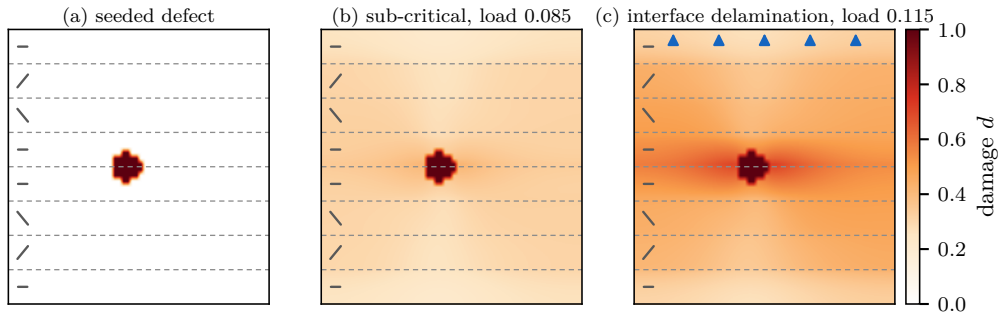
Model (12/12 tests green):

- multi-ply: per-ply $A(\theta_{\text{ply}})$, horizontal bands
- interface bands $G_c^{\text{int}} / G_c^{\text{ply}} = 0.4$ (DCB/ENF-anchored)
— pure-PF surrogate of Paggi–Reinoso CZM
- Carrara (2020) fatigue: history $\bar{\alpha}$ accumulates ψ_e^+ over cycles, $f(\bar{\alpha})$ degrades $G_c \Rightarrow$ Wöhler-like growth under *repeated sub-critical* loads

Verified physics:

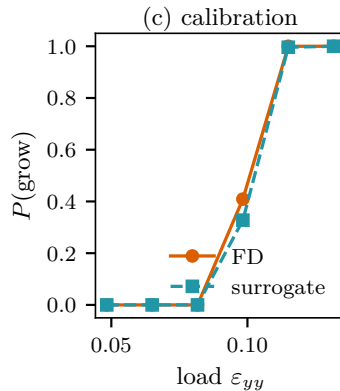
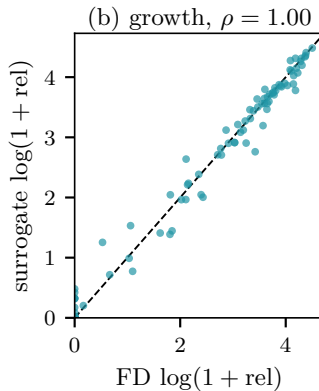
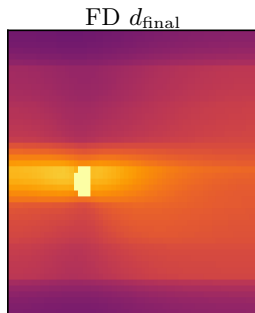
- fiber at $30^\circ \rightarrow$ crack deflects to 23.3°
- low- G_c interface captures crack (73 vs 0 new cells)
- $P(\text{growth})$ monotone in load: $0 \rightarrow \frac{2}{3} \rightarrow 1$
- FD, plane-strain, 60×48 grid, ~ 15 s/run

Caveat: load is a normalised Mode-I peel proxy; $\bar{\alpha}_T$ not yet S–N-anchored.



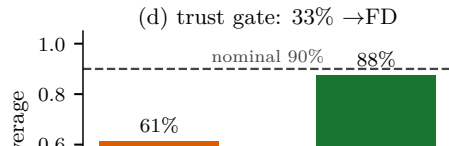
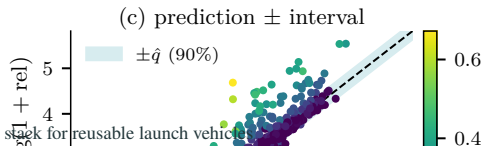
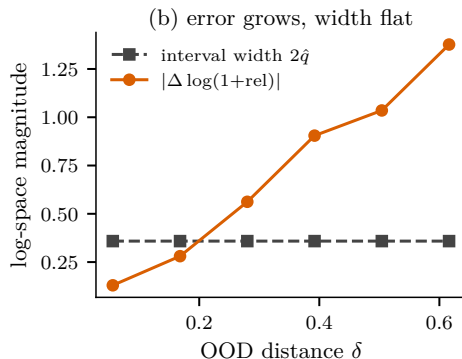
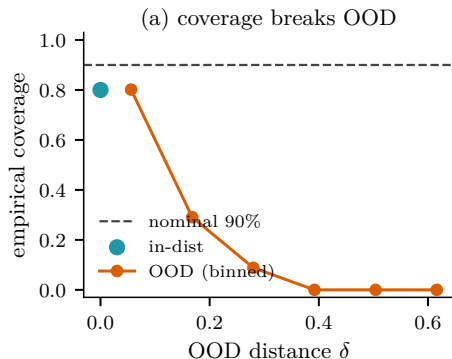
- (a) initial damage seeded at the FMPE posterior mean (c_x, c_y , layer, size); dashed lines = ply interfaces, glyphs = fiber angles ($0/\pm 30$)_s
- (b) below critical load: blunted, no propagation
- (c) above: the crack **selects the weak interface path** (delamination), exactly the failure mode DCB/ENF data anchor ($G_c^{int}/G_c^{ply}=0.4$); arrows = transverse tension

(a) load = 0.140



surrogate d_{final}

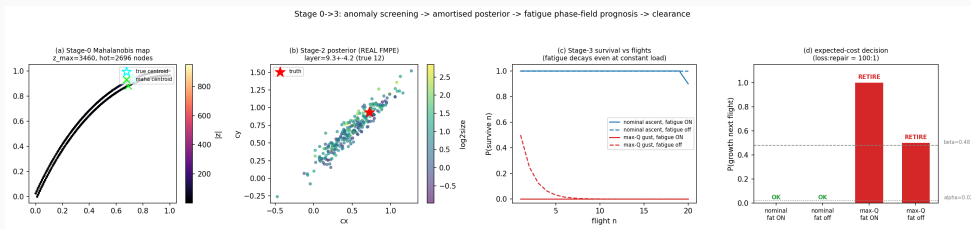
1762.9 ms(d) speed: 142× faster



Stage 0→3 run: held-out 4x4 field → Mahalanobis map → *trained* FMPE posterior → fatigue-aware clearance (20-flight horizon).

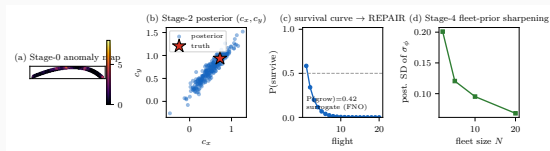
load case	load	fatigue	$P(\text{grow})$	$\mathbb{E}[\text{flights}]$	decision
nominal ascent	0.06	ON	0.00	19.9	OK
nominal ascent	0.06	off	0.00	11.0	OK
max-Q gust	0.10	ON	1.00	0.0	RETIRE
max-Q gust	0.10	off	0.50	1.0	RETIRE

- fatigue *accumulates across flights* — $P(\text{survive } 20)$ now decays even at constant sub-critical load (no i.i.d. shortcut)
- thresholds from expected cost (vehicle:repair = 100:1 $\Rightarrow \alpha=0.02, \beta=0.48$)



Stage 0 anomaly map $\cdot$ Stage 2 posterior $\cdot P(\text{survive } n)$ with fatigue $\cdot$ decision — single held-out CFRP field.

One command, four stages, under four seconds

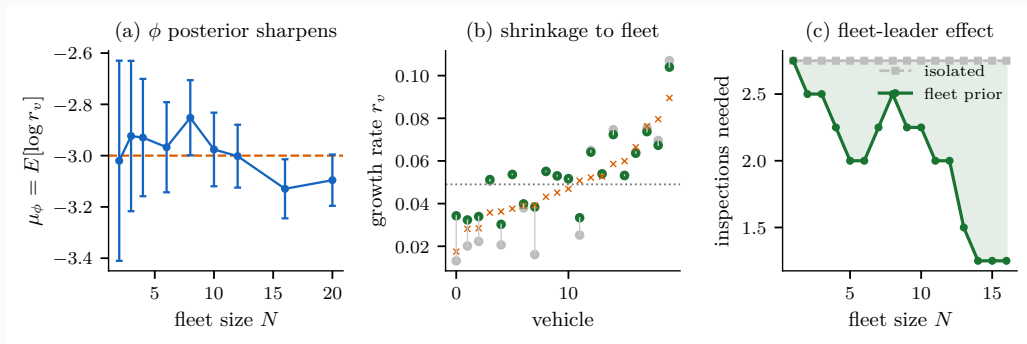


```
$ python run_pipeline.py -sample 3
```

```
[0 DETECT] AUROC 0.998  $\rightarrow$  DEFECT
[2 CHARAC] FMPE cx 0.50 $\pm$ 0.27 ...
[3 PROGNO] P(grow)=0.42
            E[remain]=1.4 (FNO 37ms/draw)
[DECISION]  $\rightarrow$  REPAIR
[4 FLEET ] prior SD 0.20 $\rightarrow$ 0.07
```

- every stage *imported unchanged*; surrogate engine $\Rightarrow$ whole run **<4s** (FD engine: Stage 3 alone $\sim$ 10 s)
- one reproducible artifact = paper + live demo + Keio proposal

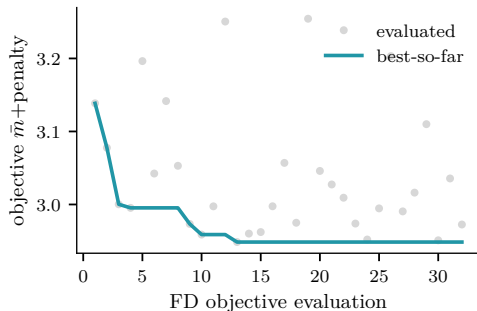
Stage 4 — fleet learning: the SpaceX loop, formalised



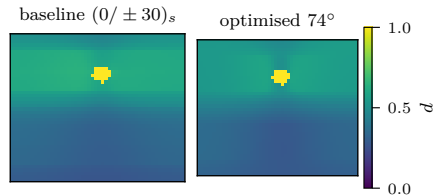
- hierarchical Bayes: fleet prior $\varphi \rightarrow$ per-vehicle growth rate $s_v \rightarrow$ inspection posteriors; conjugate Gibbs
- (a) φ sharpens with flights (SD 0.39 $\rightarrow$ 0.10); (b) low-data vehicles shrink toward the fleet mean (error ratio 0.84)
- (c) a new vehicle clears in 1.25 inspections with the fleet prior vs 2.75 in isolation — the fleet-leader effect, quantified

Stage-5 design feedback: fleet-robust laminate optimisation (exact FD phase-field forward)

(a) Bayesian-optimisation convergence



(b) worst-case defect d_{final}



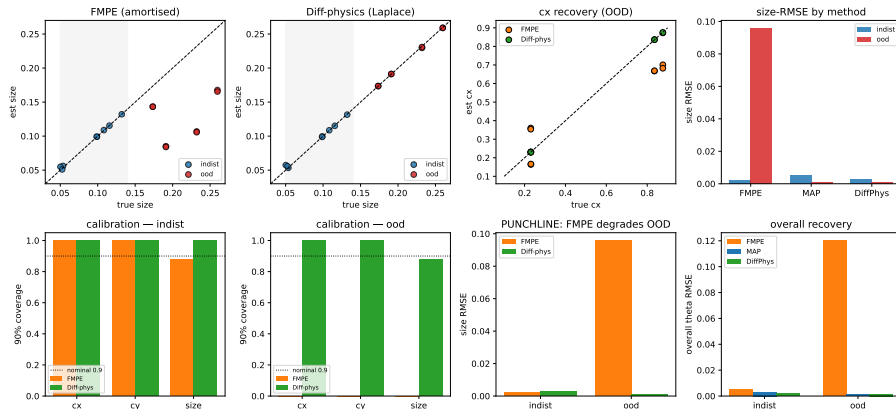
(c) robust growth (−10% mean, −2% tail)



(d) objective vs off-axis angle



Concept B — Differentiable phase-field inversion vs amortised FMPE (in-distribution vs OOD)



- JAX-differentiable AT2 forward $F(\theta)$; gradient verified to rel. 3×10^{-7} ; Laplace posterior from $J = \partial F / \partial \theta$, *no labelled pairs*

Bayesian SHM stack for reusable launch vehicles

- OOD: FMPE degrades 25x coverage $0.96 \rightarrow 0.00$: the physics-grounded posterior is invariant (C RMSE

- **Stage 4 — fleet learning:** hierarchical Bayes over tail numbers; formalises the SpaceX fleet-leader practice (lead vehicle = likelihood, fleet = prior update)
- **Stage 5 — design feedback:** fleet damage statistics → layup/reinforcement optimisation (BO)
- **Sim-to-real:** flow-matching bridge ready for first measured full-field data (per-sample normalisation needs no healthy reference)
- **Targets:** JAXA reusable-vehicle programmes; journal paper (anomaly benchmark + calibrated FMPE + fatigue prognosis)

One sentence

SpaceX learned this loop empirically over a decade — we formalise it so it works from the first inspection of the first article.